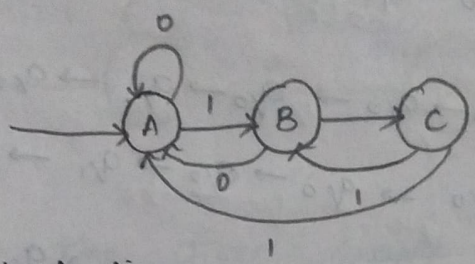
CO1 - Computational Problems.

CO1 - AT1.

Q1. How can every non-deterministic finite automation (NFA) be converted into an equivalent deterministic finite Automation (DFA) and what are the challenges involved? Consider the NFA given below ~~using~~ which accepts some language L over the alphabet $\Sigma = (0, 1)$. Construct a DFA that represents the same language L.



Introduction:

Every non-deterministic finite Automation (NFA) can be converted into an equivalent deterministic finite Automation (DFA) using the subset construction Algorithm (or powerset construction)

Given NFA:

States:

$Q = \{A, B, C\}$

Alphabet:

$\Sigma = \{0, 1\}$

Initial state : A

Final state : C

From the given NFA:

$A \xrightarrow{*} A$

$A \xrightarrow{1} B$

$B \xrightarrow{0} A, C$

$C \longrightarrow A, B$

Each DFA state represents a set of NFA states set:

$q_0 = A$ $\quad q_d = \phi$

$q_1 = B$

$q_2 = A, C$

$q_3 = A, B$

Since C is the final state of the NFA, any DFA state containing C is a final state

therefore,

$$F_{DFA} = \{\{A, C\}\}$$

Transition Calculation:

From A,

For input 0 : $S(\{A\}, 0) = \{A\}$

For input 1 : $S(\{A\}, 1) = \{B\}$

From B,

For input 0 : $S(\{B\}, 0) = \{A, C\}$

For input 1 : $S(\{B\}, 1) = \phi$

From A, C :

For input 0 : $S(\{A, C\}, 0) = \{A\}$

For input 1 : $S(\{A, C\}, 1) = \{A, B\}$.

From A, B :

For input 0 : $S(\{A, B\}, 0) = \{A, C\}$

For input 1 : $S(\{A, B\}, 1) = \{B\}$

From ϕ

There are no positive transition, so:

$S(\phi, 0) = \phi$

$S(\phi, 1) = \phi$

• DFA Transition Table

DFA	Input 0	Input 1	Accepting ?
→ [A]	[A]	[B]	No
[B]	[C]	[ϕ]	No
* [C]	[A]	[A, B]	Yes
[A, B]	[A, C]	[B]	No
* [A, C]	[A]	[A, B]	Yes
[ϕ]	ϕ	[ϕ]	No.

Conclusion:

Using the subset Construction method, the given NFA is converted into an equivalence DFA. The DFA has the same accepted language L, while providing exactly one transition for each input symbol from every state.

Q2. How is mathematically induction used to prove properties of strings and languages in automata theory? Apply the mathematical induction and prove that for all positive integers n:

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

Mathematical induction is a method used to prove that a statement is true for every positive integer n.

Take : $n = 1$

$$\sum_{i=1}^{1} i^2 = 1^2 = 1$$

R.H.S:

$$\frac{1(1+1)(2(1)+1)}{6} = \frac{1 \times 2 \times 3}{6} = 1$$

$\therefore$ LHS = RHS.

Assume that the statements is true for $n = K$

Therefore,

$$\sum_{i=1}^{K} i^2 = \frac{K(K+1)(2K+1)}{6}$$

This is called Induction Hypothesis.

We need to prove that the statement is true for

$$n = K + 1$$

$$\sum_{i=1}^{K+1} i^2 = 1^2 + 2^2 + \ldots\ldots K^2 + (K+1)^2$$

Using inductive hypothesis

$$= \frac{K(K+1)(2K+1)}{6} + (K+1)^2$$

$$= (K+1)\left[\frac{K(2K+1)}{6} + (K+1)\right]$$

$$= \frac{(K+1)[K(2K+1) + 6(K+1)]}{6}$$

$$= \frac{(K+1)(2K^2 + 7K + 6)}{2}$$

This does not match the required formula

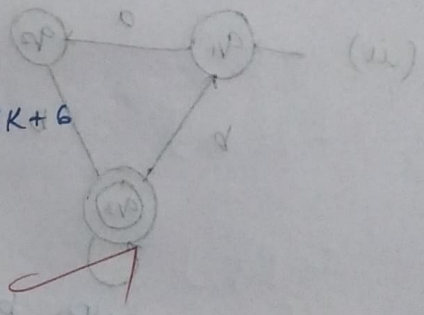
$$\frac{K(K+1)(2K+1)}{6} + (K+1)^2$$

$$= \frac{K(K+1)(2K+1) + 6(K+1)^2}{6}$$

$2K^2 + 7K + 6$ is inconsistent

$\therefore (K(2K+1) + 6(K+1)) = 2K^2 + 7K + 6$

which factors as $(2K+3)(K+2)$.



Now, Since

$$(K+1) + 1 = K+2$$

and

$$2(K+1)+1 = 2K+3$$

we get, $\frac{(K+1)((K+1)+1)(2(K+1)+1)}{6}$

Thus the statements holds for $K+1$.

Conclusion:

The statement is true for ~~n~~ $= 1$ and if it is true for $n = K$; it is also true for $n = K+1$

$\therefore$ By principle of mathematical induction

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

is true for all positive integers